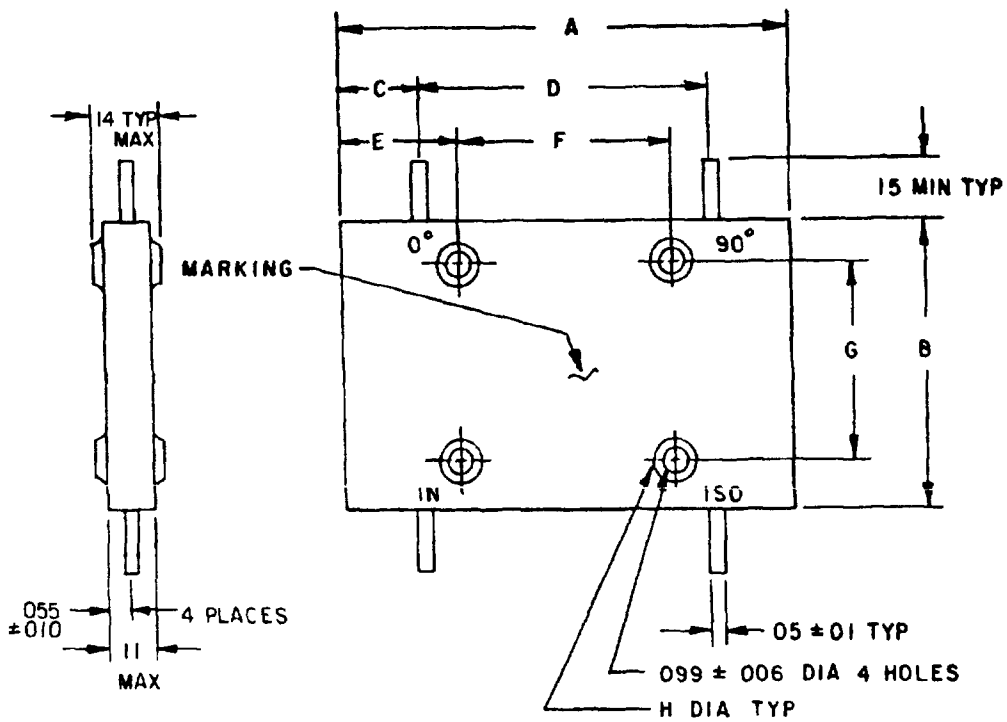
MILITARY SPECIFICATION SHEET

POWER DIVIDER/COMBINERS, QUADRATURE, STRIPLINE

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the power divider/combiners described herein shall consist of this document and the latest issue of specification MIL-P-23971.



INCHES	MM
.006	.15
.01	.25
.010	.254
.05	1.27
.055	1.40
.099	2.51
.11	2.79
.14	3.56
.15	3.81

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

FIGURE 1 Dimensions and configuration dash numbers 01 and 02

FSC 5985

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Table I Dimensions.

M23971/12-	A	B	C	D	E	F	G	H
01	1.83 ±.02	2.02 ± 02	18 ± 02	1.460 ± 015	37 ±.02	1.060 ±.015	1.750 ±.015	.22 max
02	1 00 ±.01	87 ±.01	17 ± 01	66 ± 01	343 ± 010	313 ± 010	.625 ±.010	---

INCHES	MM	INCHES	MM
.01	.2	625	15.88
010	.25	.66	16.76
.015	.38	87	22.10
.02	.51	1 00	25 40
.17	4.32	1 060	26.92
.18	4.57	1.460	37.08
.22	5.59	1.750	44.45
.313	7.95	1.83	46.48
.343	8.71	2.02	51.31
.37	9.40		

TABLE II Electrical performance characteristics and physical requirements.

M23971/12-	Impedance (ohms)	Frequency range (GHz)	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance max (degree)	Amplitude balance (dB)	Power level avg (W)	Power level pk (W)	Weight pounds (grams)	Ambient temperature Operating	Ambient temperature Storage	Figure
01	50	.1-.16	3. +.2 -.0	1.2:1	0.3	25	±2.0	±0.6	200	1K	.038 (18)	-55° to +71°C	-55° to +65°C	1
02	50	.5-1.0	3. +.2 -.0	1.2:1	0.25	20	±1.5	±0.5	200	1K	.054 (25)	-20° to +75°C	-55° to +85°C	1

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REQUIREMENTS

Design and construction See figure 1

Housing Hermetically sealed

Material
Dash numbers 01 and 02 6061-T6 aluminum

Finish
Dash number 01 Clean irridite finish per MIL-C-5541, type 1, grade C, class III
Dash number 02 Gold plated per MIL-G-45204, type 1, grade C, 000050 to 000100 inch thick over 2-ounce copper clad microstrip

Terminals

Material
Dash number 01 Nickel alloy
Dash number 02 Silphos (15% silver, 80% copper, 5% phosphorus)

Finish
Dash number 01 tin plated
Dash number 02 Nickel clad, 004 inch thick nickel plus 001 thick silphos
Solder dipped with Sn60 or Sn63 per QQ-S-571

Electrical characteristics See table II

Weight See table II

Ambient temperature See table II

Environmental tests In accordance with MIL-P-23971 except
Dash number 02
Vibration, high frequency in accordance with method 204 of MIL-STD-202, test condition B

Part number M23971/12-(dash number from table II)

NOTE Power dividers covered by this specification sheet are substituted for the following manufacturer's part number This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number

Dash number	Manufacturer's code	Manufacturer's part number	FSN
01	00752 31597 12457 31550	406118-1 1A0280-3 QH9021348 504736-2	—
02	31597	10264-3	—

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force -
DLA - ES

User activities:
Army -
Navy - AS, OS, MC
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

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DOCUMENT IDENTIFIER AND TITLE MIL-P-23971/13

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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